Responses to editor and reviewer concerns

Thank you for the opportunity to revise and resubmit our manuscript, "Improving Webcam Eye-Tracking: Impact of Hardware and Head Stabilization on Data Quality" (Manuscript ID BR-Org-25-1067). We found the reviewers' comments to be insightful and constructive, and we have used them to substantially revise and strengthen the manuscript. In this memo, we describe how we addressed each point. Reviewers' comments are cited or summarized in red italicized text, and our responses appear in black Roman text. Excerpts from the revised manuscript are shown in blue. All added or modified text in the manuscript is indicated in bold. We have included both a marked-up and a clean version of the revised manuscript.

Both reviewers offered thoughtful suggestions regarding alternative experimental manipulations. We greatly appreciated these ideas and agree that multiple additional factors could be investigated to further improve webcam-based eye-tracking. For the present Stage 1 Registered Report, however, we elected to retain the originally proposed manipulations in order to maintain a focused and interpretable experimental design.

Importantly, both reviewers acknowledged the merit and relevance of the current manipulations for addressing our primary research questions. Expanding the design at this stage would substantially increase complexity and reduce the clarity of the planned comparisons. We therefore propose to proceed with the current design as specified, while incorporating the reviewers' suggestions into the theoretical framing and discussion. We view the present study as a foundational step rather than an exhaustive evaluation of all methodological factors influencing webcam-based eye-tracking. We hope the results will motivate future work—by us and others—examining additional forms of head stabilization and platform-specific implementations.

The major revisions to the Registered Report are as follows:

> **!** Important
>
> - Introduction: we reorganized the introduction and added discussion on lab based work using webgazer.js
> - Stimuli: We now restrict the analyses to TCUU trials only.
> - Analysis plan: We adopt a binarized (winner-take-all) approach and focus on the unrelated item that is not phonologically related to the cohort.
> - Sampling plan: We re-ran the power analysis using an intercept-only binomial count model based on the binarized outcome. Under this model, the planned sample size provides adequate power to detect a small cohort effect.

Reviewer 1

Major Concern

> **Reviewer 1**: *While these experiments are expected to provide clear answers to the research questions, I think they are likely to have a minimal impact on how researchers conduct web-based*

> *eye-tracking studies. The authors note that previous studies suggest clear relationships between webcam quality and calibration accuracy, and between head movement and calibration accuracy, so the planned experiments would be largely confirmatory in nature. Additionally, the preference for using a higher-quality webcam and a chin rest already exists, and empirically showing effects of webcam quality and the use of a chin rest is unlikely to change methodological considerations of the researchers.*

We respectfully disagree with the reviewer's assessment that the present work will have minimal impact on how researchers conduct web-based eye-tracking studies. Although many webcam eye-tracking studies are conducted "in the wild," this methodology is increasingly being adopted for controlled laboratory use, particularly by researchers for whom access to dedicated eye-tracking hardware is limited by cost or infrastructure. In such contexts, concrete guidance about which webcams may need to be excluded and how sensitive results are to head movement is highly consequential for study design and data quality.

While prior work has noted associations between webcam quality, head movement, and calibration accuracy, these relationships have largely been observational or descriptive in nature. The present study goes beyond this by experimentally testing how these factors impact eye-movement data quality and downstream measures, thereby allowing stronger inferences about their practical importance. This distinction is critical for informing methodological decisions, such as whether a virtual chin rest meaningfully improves data quality or whether certain classes of webcams should be avoided altogether.

As such, we view the contribution of this work not as merely confirmatory, but as providing empirical constraints on methodological choices that are currently guided largely by intuition, convention, or anecdotal best practices. We believe these findings will contribute to specific, empirically grounded recommendations that can optimize webcam set up for both researchers conducting web-based eye tracking and those seeking cost-effective alternatives within the laboratory setting.

> **Reviewer 1***: The authors may wish to examine the effects of other factors that have a larger impact on researchers' methodological decisions or instructions given to participants. For example, how eye-movement data quality is affected by the relative position of a ceiling light or a desk lamp, or the relative height of the laptop. Answers to these questions will be informative as there currently seem to be no standard instructions about these aspects in webcam eye-tracking studies.*

We thank the reviewer for this thoughtful suggestion. We agree that factors such as ambient lighting conditions (e.g., ceiling lights vs. desk lamps) and device positioning (e.g., laptop height) are likely to influence eye-movement data quality in webcam-based eye tracking. In the proposed work, due to practical constraints related to time and resources, we plan to keep lighting conditions fixed to minimize interaction effects between hardware variables and lighting. However, we view lighting variables as an important direction for future research and hope that subsequent work will be able to more comprehensively examine how such environmental and setup variables affect data quality and inform the development of standardized participant instructions.

Minor

> **Reviewer 1**: *p. 5: "if (H2a)" Is this a typo?*

Thanks for spotting this. It should just read "(H1a)". We have made this change.

> **Reviewer 1**: *p. 7: "Eye movements are recorded continuously during each ." The final word appears to be missing.*

We have edited this sentence.

> Eye movements are recorded continuously during the final image display. After the main task, participants complete a brief demographic questionnaire and are thanked for their participation.

> **Reviewer 1**: *p. 8: "Concretely, For each trial…" -> "Concretely, for each trial…"*

We have corrected this.

Reviewer 2

> **Reviewer 3**: *Suggestion 1: Head stabilisation manipulation Experiment 2 The suggestion that would lead to the most substantial changes concerns Experiment 2. Although I see the merit of the chin rest manipulation, Experiment 2 might have greater impact if head stabilisation were manipulated in a way that better reflects remote testing, which is the most common use of webcam eye-tracking. In particular, the authors note in the introduction to Experiment 2 (p. 14) that platforms like labvanced can warn participants whenever they move their heads too much. I think that the impact of Experiment 2 would be stronger if head stabilisation were manipulated as a "warning" vs. "non-warning" condition, rather than using a chin rest. This approach would not only test the effect of head stabilisation, but now in a way that it could potentially offer a practical, off-the-shelf solution for researchers that will use webcam eye-tracking in remote settings.*

We thank the reviewer for this constructive suggestion and agree that a manipulation based on real-time head-movement warnings, as implemented in platforms such as Labvanced, would be highly relevant for fully remote webcam eye-tracking studies.

For the present Stage 1 Registered Report, however, we elected to manipulate head stabilisation using a chin rest in a controlled laboratory setting. This choice was made at the design stage to ensure precise experimental control over head movement while holding all other aspects of the task and platform constant. At present, the experimental platform used in this study (Gorilla) does not provide real-time head-tracking or automated movement warnings, and comparable functionality is not natively available in widely used open-source platforms such as jsPsych or PsychoPy (although this can be coded). As a result, implementing a warning-based manipulation would require a different platform with proprietary tracking methods, introducing additional sources of variability that would complicate interpretation at this preregistration stage.

Importantly, the goal of Experiment 2 is not to evaluate a specific commercial implementation of head-movement warnings, but to test the causal role of head stabilisation on calibration quality and gaze precision in webcam eye-tracking. The chin-rest manipulation provides a principled and conservative test of this mechanism by directly constraining head movement. The resulting estimates can therefore serve as an upper bound on the benefits that any form of head-stabilisation —whether via participant instruction, automated warnings, or future software-based solutions— might yield in both laboratory and remote contexts.

We now clarify this rationale in the manuscript and explicitly frame the chin-rest manipulation as a theoretically motivated proxy for head-stabilisation instructions and emerging software-based approaches, rather than as a direct analogue of existing warning systems.

> In Experiment 2, we will use the same standard-quality external webcam as in Experiment 1, but will manipulate head stability by comparing a chin-rest condition to a no–chin-rest condition. hessels2014 offered evidence that head movement can impact the accuracy and reliability of eye-tracking estimates, showing that even moderate deviations in head position can produce systematic calibration drift, increased data loss, and slower recovery when gaze is reacquired. Importantly, their study demonstrates that these effects are not limited to extreme movements and can arise during typical participant behavior when head position is not constrained.
>
> Some online platforms, such as Labvanced [Finger2017LabVanced], attempt to mitigate head motion by warning participants when they move outside a predefined region. However, such approaches rely on proprietary head-tracking solutions and, at present, it remains unclear how warning-based motion control interacts with WebGazer.js estimates of event detection, onset latency, and gaze competition. Moreover, comparable real-time head-movement warnings are not currently available in commonly used platforms such as Gorilla, jsPsych, or PsychoPy, limiting the feasibility of implementing such a manipulation within a single preregistered design.
>
> In laboratory-based eye-tracking, chin rests are routinely used to stabilize the head in a fixed position so that only the eyes move during the experiment. Although chin rests are uncommon in fully remote testing, introducing a chin-rest manipulation provides a controlled and conservative test of the causal role of head stability in webcam-based eye-tracking. Specifically, the chin-rest condition serves as a theoretically motivated proxy for head-stabilisation instructions and emerging software-based warning systems, and allows us to estimate an upper bound on the benefits that head-movement control may provide for calibration quality, competition effects, event-detection timing, and participant attrition. We therefore test the following hypotheses regarding competition, onset, and attrition.

> **Reviewer 2**: *Suggestion 2: Omit the rhyme picture from the experimental display I was surprised that the experimental display included a rhyme competitor, which is (as I understand it) not directly related to the hypotheses or analysed in the results. While I understand that any effects of the rhyme competitor would likely impact both webcam conditions equally (and thus not affect the main conclusions), I wonder why this rhyme competitor is necessary. In fact, it might be preferable to replace the rhyme picture with an unrelated one. If anything, including a rhyme competitor might reduce looks to the target and perhaps even the cohort (albeit at a later, theoretically less relevant timepoint in the trial), which would unnecessarily shrink the effect of interest (i.e., the competitor effect at the word onset). Considering this, I would recommend the authors to either replace the rhyme pictures, or describe why including this picture is relevant to the hypotheses/analyses.*

We thank the reviewer for this suggestion and agree that including a rhyme competitor is not necessary for the hypotheses or analyses in the present study. The inclusion of rhyme competitors in earlier versions of the task was based on convenience and consistency with a prior validation study conducted during the development of the {webgazeR} package, rather than theoretical necessity for the current work. Because rhyme competitors were not relevant to our hypotheses and could plausibly dilute early target–cohort competition effects, we have revised the design to omit rhyme images entirely. In the revised experiment, all trials are of the TCUU type, containing one target, one onset cohort competitor, and two unrelated images. This change more directly isolates the competitor effect of interest at word onset.

> Stimuli were adapted from colby2023. Each stimulus set comprised four images. For the webcam study, we used 60 stimulus sets (30 monosyllabic, 30 bisyllabic). All trials were of the TCUU type (target–cohort–unrelated–unrelated), in which a target, its onset cohort competitor, and two unrelated images were displayed (e.g., MOUTH, mouse, house, chain). This design yielded 60 trials in total, with each stimulus set contributing one trial. A custom MATLAB script (R2024a; https://osf.io/x3frv) generated a unique randomized trial list for each participant, pseudo-randomizing display positions such that the target, cohort, and unrelated images were approximately equally likely to appear in each quadrant across participants.

> **Reviewer 2**: *Suggestion 3: Calibration and attrition The authors expect that attrition rates are lower in the high-quality webcam condition. This is interesting, but the way in which the hypotheses and analyses related to this expectation are written up was somewhat confusing. In particular, the hypothesis related to attrition is described in terms of attrition (p.5), but its associated analysis is described in terms of calibration (p. 13). Does this imply that the experiment stopped for participants who failed to calibrate? Moreover, I thought the description of the calibration scores was a bit unclear. How are these scores analysed? And are the calibration scores described on p. 8 (which are values between 0 to 1, binarised with a cut-off at 0.5) the same as the calibration scores included in the analyses described on 0. 13? I would recommend the authors to create a (sub)section in which the calibration procedure and its scores are explained in more detail.*

We are sorry this was not made clearer. Attrition is tied to calibration. That is, if individuals fail calibration they are not allowed to continue (if they passed the first calibration but not the second) or start the task (failed the first calibration task).

We have made this clearer throughout the manuscript and have also added a subsection with more details about the calibration as requested.

> **Webcam Calibration**
>
> To ensure that participants understood how webcam-based eye tracking works, they first viewed an instructional video demonstrating the calibration procedure (available on OSF here: [link]). Calibration was completed twice—once at the beginning of the experiment and again after 30 trials. Each calibration session allowed up to three attempts.
>
> During calibration, participants completed a passive calibration task in which nine red targets were presented sequentially on the screen. Participants were instructed to look directly at each target while it was visible. We used default system parameters for the calibration procedure,

> including the required fixation duration per target (200 ms), the number of gaze samples used per prediction (10 prediction points), and the transition time before data collection began (1000 ms).
>
> Immediately following calibration, validation was performed using five green targets presented one at a time. Participants were again instructed to look directly at each target. During validation, the eye-tracking system evaluated calibration quality by comparing the predicted gaze location to the known target location. Calibration was considered unsuccessful if more than two validation points deviated from their intended target location (i.e., predicted gaze was misaligned with another calibration point).
>
> Participants who failed any calibration check were branched to the end of the experiment and asked to complete a brief demographic questionnaire.

Confidence and convergence are not tied to calibration but to the accuracy model predictions. We added more information about these metrics here

> In addition, Gorilla provides two eye-tracking quality metrics derived from the underlying face-tracking model: convergence and confidence. Convergence (range: 0–1) reflects the model's certainty that a face has been successfully detected, with higher values indicating poorer convergence on a face. Confidence (range: 0–1) reflects the support vector machine (SVM) classifier's confidence in the detected face. Trials will be excluded if convergence exceeds 0.5 (indicating unreliable face detection) or if confidence falls below 0.5 (indicating low classifier certainty).

> **Reviewer 2**: *Suggestion 4: Network control The authors argue that the experiments will be carried out with "controlled network settings". This is very important, because internet speed can affect webcam performance a great deal even with high-end webcams (think of freezes in a videocall). Therefore, I'm happy to hear that the authors take this into account, but I would like to know more details about how the network settings will be controlled. Will the authors use some procedure to ensure that internet speed is stabilised for all experimental sessions? Or, is the network controlled insofar that all participants take part from the same computer and network? In case of the latter, I would recommend to do a short internet speed check prior to each experimental session, because internet speed can still vary. Also, I would recommend not only controlling the internet speech, but also the browser speed as this might affect the speed in which WebGazer's functions are executed. Because all participants will use the same browser (version) and hardware, I think that making sure to clear the browser cache before each experimental session could already be enough.*

Thank you for pointing this out and excellent suggestions. It is important to adequately describe this in a bit more detail. Computers will be hardwired into the campus network. In addition, per your suggestion, we will clear the browser cache before each experimental session and also run a speed check on each computer.

> **Reviewer 2**: *On p.3 (line 34), the authors argue that "no study has directly tested how environmental and hardware constraints impact webcam-based eye-tracking data". Like I already described at the start of this review, this is not entirely true, as Semmelmann & Weigelt (2018) and Slim, Kandel et al., (2024) have also provided insights into this matter.*

We have added a discussion of Semmelamnn & Weigelt and Silm et al in relation to lab-based use of webgazer.js.

Several studies have examined the performance of WebGazer-based eye-tracking in controlled laboratory settings relative to online data collection, with mixed results. For example, semmelmann2018 reported reduced spatial noise when WebGazer was used in the laboratory compared to online settings, whereas slim2024 found largely comparable effects between lab-based and remote samples. Importantly, however, these comparisons primarily captured broad differences between controlled and uncontrolled testing contexts and were not designed to isolate the effects of specific hardware characteristics or environmental constraints.

More targeted evidence suggests that hardware quality may nonetheless contribute to variability in webcam-based eye-tracking data. For instance, slim2023 reported a positive association between sampling rate and calibration accuracy, and gellerLanguageBordersStepbystep2025f found that participants who failed calibration more frequently were more likely to report using standard-quality built-in webcams and completing the study in suboptimal environments (e.g., natural lighting). To date, no study has directly examined the impact of head stabilization on webcam-based eye-tracking, despite extensive evidence from laboratory eye-tracking with research-grade eye-trackers showing that head movement can substantially reduce both spatial accuracy and precision [hessels2014]. Together, this body of work suggests that while general hardware variation may exert relatively small effects on webcam-based eye-tracking outcomes, the isolated contributions of webcam quality and participant stability remain poorly understood. Accordingly, in the present research we directly manipulate two factors—hardware selection (webcam quality) and participant stability—across two experiments to more precisely characterize their impact on data quality.

Reviewer 2*: On page 2, it's described that webcam-based eye-tracking is implemented in several online experiment platforms like PsychoPy, jsPsych, Gorilla, or PCIbex. It could be made explicit that all these platforms involve some version of the WebGazer algorithm, which is introduced several paragraphs later. Perhaps a more logical order of this section would be to first introduce the WebGazer algorithm, and then describe that this algorithm has been implemented in several popular experiment platforms.*

Thank you for this helpful suggestion. We have revised the organization of the Introduction to first introduce the WebGazer algorithm and then describe its implementation across commonly used experimental platforms (e.g., PsychoPy, jsPsych, Gorilla, and PCIbex). We believe this reordering improves the logical flow and clarity of the section.

Reviewer 2*: The (general) introduction of the paper discusses the potential impact of hardware/ webcam quality on webcam-based eye-tracking, while the potential impact of*

head stabilisation is described more in-depth in the specific introduction of Experiment 2 (p. 14). Given that head stabilisation is the main motivation for Experiment 2, a discussion of its (potential) effects warrants a more prominent place in the general introduction as well.

Thank you for this helpful suggestion. We have revised the general Introduction to include a more explicit discussion of the potential role of head stabilization in webcam-based eye-tracking. In the revised manuscript, we now introduce head stabilization alongside hardware considerations in the general Introduction, reflecting its central role as the motivation for Experiment 2, and clarifying the rationale for manipulating hardware selection and participant stability across the two experiments.

> **Reviewer 2**: *The authors determined their sample size based on a power analysis. I was wondering whether this meant that participants that are excluded will be replaced (I think the answer to this is "yes", because the authors describe that they will run the study until they have 100 "usable" participants, but it would be clearer if this were described explicitly).*

This is correct. We have updated our sampling plan section to make this clearer.

> Data collection will continue until 100 usable participants are obtained (50 per webcam-quality group). That is, participants will be replaced if they fail calibration at any point. For analyses of attrition (i.e., failed calibration attempts) (see below), all participants who enter the study will be included, thus we might have more than 100 participants total.

> **Reviewer 2**: *The authors are planning to aggregate the data into 50 ms time bins. Although I understand that such 50 ms time bins provide quite some temporal precision in the analyses, I wonder whether it would be more realistic to aggregate into 100 ms time bins. In my experience, WebGazer's sampling resolution is quite inconsistent within participants – and even within trials. Therefore, I worry whether you might end up with a relatively large number of empty time bins (especially in the standard-quality webcam condition).*

Thank you for this thoughtful comment. We agree that sampling irregularity is an important consideration when selecting temporal bin widths for webcam-based eye-tracking data, particularly given known variability in WebGazer's effective sampling rate both within and across participants.

Our initial motivation for using 50 ms time bins was driven by the planned GAMM analyses, which benefit from higher temporal resolution when modeling fine-grained time-course effects. Because data collection for the present study occurs in a controlled laboratory setting, we expect more stable sampling than is typical for fully remote webcam studies. In the revised protocol, we therefore clarify that participants with effective sampling rates below 30 Hz will be excluded, rather than the previously stated 15 Hz threshold. Under these conditions, 50 ms bins should generally contain at least one sample per bin, even in the standard-quality webcam condition.

At the same time, we recognize the reviewer's concern that sampling irregularities may still lead to empty or sparsely populated bins, particularly for lower-quality webcams. To address this, we have added a contingency plan to the manuscript under footnote 2:

> If inspection of the data reveals that each participant × trial × time bin does not contain at least one valid sample, we will aggregate the data into 100 ms time bins and relax the 30 Hz sampling rate requirement (use 15 Hz instead). This approach provides a more conservative temporal resolution while ensuring robust parameter estimation and minimizing unnecessary data loss.

Together, this strategy balances the analytic advantages of finer temporal resolution with the practical constraints of webcam-based eye-tracking and ensures that our conclusions are not driven by artifacts of sampling variability.

Reviewer 2: *The description of the dependent variable was not very clear to me (i.e., "gaze counts to cohorts compared to unrelated items"). What is meant with "gaze counts" here? Is it the number of frames (within a time bin) in which WebGazer detected a look to the cohort item minus the number of frames (within a timebin) in which WebGazer detected a look to an unrelated item? In general, I would suggest binarising dependent variables in eye-tracking data: If a participant looks at a specific region of interest (e.g., the cohort) within a given time bin, the fixation to that region could be coded as '1' for that time bin. This conceptualisation of the dependent variable more closely reflects the nature of gaze behaviour (i.e., you either look at something or you don't). That said, my expertise with GAMMs is limited, so there may be model-specific reasons for implementing the dependent variable differently that I am not aware of.*

We have made our analysis plan a bit clearer here and we are indeed binarising our variable as you suggested.

> Before fitting the GAMM, the eye-tracking data were processed to obtain a binomial response suitable for model fitting. Gaze samples were first binned into 50-ms intervals within each trial.1 For each participant, trial, and time bin, we computed the number of valid gaze samples ("looks") directed to the cohort image and to the unrelated image. For the present analysis, we included only the unrelated competitor that was fully unrelated to the target. A second competitor, which was phonologically related to the cohort but not to the target, was excluded from this analysis. Bins containing no looks to either relevant image were excluded from further analysis. Each remaining bin was then binarized using a winner-take-all procedure: a bin was coded as 1 if the cohort image received more looks than the unrelated competitor in that bin, and 0 otherwise. This binarization yields a single binary observation per time bin within each trial, effectively downsampling the gaze-sample stream while preserving time-resolved preference dynamics. We refer to these 0/1 observations as looks (rather than fixations) to emphasize that the dependent measure is defined over gaze samples rather than discrete oculomotor events. After binarization, trial-level 0/1 looks were aggregated across trials to obtain binomial counts for each participant × condition × time-bin combination: the number of trials coded as 1 (cohort_looks) and the number of trials contributing valid data (total_looks). These binomial counts (cohort_looks out of total_looks) constituted the response variable in the GAMM. Although cohort_looks / total_looks can be interpreted descriptively as the proportion of cohort-dominant bins, proportions were not precomputed for model fitting. Instead, the GAMM was fit directly to the binomial counts on a latent log-odds scale, and time-course estimates were obtained as predicted probabilities from the fitted model. This simplified analysis approach offers several advantages. First, aggregating gaze data into binomial outcomes substantially reduces data dimensionality and attenuates the extreme temporal autocorrelation that is common in high-frequency eye-tracking data . Rather than enabling autocorrelation to be modeled directly, aggregation makes the remaining autocorrelation more tractable. Second, this aggregation strategy permits the use of more parsimonious models that are faster to fit, less computationally intensive to simulate from, and less prone to overfitting.

Reviewer 2: *On p.3 (first sentence), the authors argue that webcams with a low sampling rate might negatively affect the temporal resolution of the data. While this seems plausible, I think that sampling rate only has a weak influence on temporal delays in VWP experiments at best. With a sampling rate of 30 Hz, there are approximately 33 ms in-between frames. In the VWP, it* *typically takes around 200 ms after the onset of an auditory stimulus for participants to fixate the corresponding object. Even at 30 Hz, the temporal resolution should therefore be sufficient to detect early looks to the target. Given this, I would appreciate some further clarification on how the authors conceptualise the link between sampling rates and temporal resolution of WebGazer data. o My own two cents: Temporal delays are driven by (i) WebGazer's execution time (which, as the authors note on p.3, seems largely improved in newer implementations), and (ii) reduced spatial accuracy. Because effects in the VWP emerge gradually (due to averaging over participants/items), they are small at the onset. If the spatial accuracy is poor, early and subtle gaze shifts may not be detected reliably, making the effect appear delayed. In this case, the issue*

would be less due to temporal sampling and more about when gaze shifts become large enough to be spatially distinguishable.

Thank you for this comment. We agree that, at sampling rates commonly used in webcam-based eye tracking (e.g., ~30 Hz), temporal sampling frequency alone is unlikely to be the primary limiting factor for detecting early gaze shifts in the visual world paradigm especially with larger AOIs. As the reviewer notes, with inter-sample intervals of approximately 33 ms, such sampling rates are, in principle, sufficient to capture eye movements that typically occur on the order of ~200 ms following stimulus onset.

We have revised the manuscript to clarify that our concern is not that low sampling rates impose large temporal delays in a strict sense. Rather, we conceptualize the link between sampling rate and apparent temporal resolution as indirect and mediated by variability in sampling, execution-time delays, and reduced spatial accuracy. When spatial precision is degraded—whether due to webcam quality, head movement, or algorithmic constraints—early and subtle gaze shifts may not be reliably distinguishable from noise. In paradigms where effects emerge gradually and are small at onset, this can delay the point at which gaze differences become detectable at the aggregate level, giving rise to the appearance of temporally delayed effects.

Consistent with the reviewer's suggestion, we now emphasize that apparent temporal delays in WebGazer-based eye-tracking data are more plausibly driven by a combination of (i) execution timing and variability within the gaze-estimation pipeline and (ii) degraded spatial accuracy, rather than insufficient temporal sampling per se. We thank the reviewer for this clarification, which has helped us sharpen our conceptualization of temporal resolution in webcam-based eye tracking.

Reviewer 2*: In the power analysis: pilot participants were assigned to 'high' and 'standard' quality groups. How was this determined with the remote pilot participants? Did they fill in their webcam type in a questionnaire, or did you determine post-hoc based on sampling resolution?*

We apologize for the confusion. The pilot data were used solely to estimate a baseline cohort effect for the purpose of seeding the simulations. Participants in the pilot study were not assigned to "high" versus "standard" webcam quality groups, nor were such groups defined post hoc based on sampling resolution. In response to your recommendation, we redid the power simulation using the binarized approach. Additionally, we focused exclusively on estimating the overall cohort effect for the TCUU trials only. We did not simulate webcam quality or head stability parameters. Accordingly, the power simulations were intended only to provide guidance on the number of participants required to detect a small cohort effect (more looks to cohort), rather than to model technical variability in recording conditions. ::: {.memo-excerpt} Sampling Goal We conducted an a priori power analysis using the {simr} package in R. Data from 21 participants, collected online using the Gorilla experimental platform during development of the {webgazeR} package and employing the same stimuli and visual-world word paradigm (VWP) design, were used to seed the simulations. We focused specifically on the cohort effect in TCUU trials. Importantly, these pilot data did not manipulate webcam quality or head stabilization. Using these data, we constructed a binarized Looks variable within each time bin, thereby downsampling the gaze data. In the simulation, gaze samples were aggregated into 100-ms bins (10 Hz). Within each bin, the cohort image was coded as 1 if it received more looks than the unrelated image; otherwise, it was coded as 0. These binarized values were then aggregated across trial × time bin to obtain counts of looks to the cohort and to the unrelated image. We fit an intercept-only generalized linear mixed-effects model (GLMM) to these aggregated data. To simulate the expected effect, we specified a small effect size on the log-odds scale (b = 0.10) for the cohort versus unrelated contrast. Simulated datasets were generated under this model, and the planned GLMM was refit to each simulated dataset (5,000 simulations). Statistical power was estimated as the proportion of simulations in which the absolute z-value for the target effect exceeded 1.96. The full analysis script used to conduct this power analysis is available on OSF (https://osf.io/4trmn/files/a46g8). Although this approach simplifies the planned design, it provides a conservative lower-bound estimate of the effects expected in the current study. Results indicated that a sample size of 40 participants per group (N = 80 total) would provide approximately 90% power to detect the hypothesized cohort effect. Because this analysis focused on overall fixation proportions and we additionally plan to examine effects across time, we will recruit 50 participants per group (N = 100 total). Data collection will continue until 100 usable participants are obtained (50 per webcam-quality group). That is, participants will be replaced if they fail calibration at any point. For analyses of attrition (i.e., failed calibration attempts) (see below), all participants who enter the study will be included, thus we might have more than 100 participants total. :::

Minor

> **Reviewer 2***: I think spelling out "VWP" as "visual world paradigm" in the paper's keywords might enhance visibility of the paper.*

The keyword VWP in the abstract has been changed to the visual world paradigm.

> **Reviewer 2***: P. 5: Replacing the period (.) with a colon (':') after the sentence "We hypothesize several effects related to competition, onset, and attrition" would warn the reader that the specific hypotheses are coming up.*

This has been rewritten to be clearer.

> **Reviewer 2***: P.5: The "if" in "if (H2a) Each webcam condition will show …" seems misplaced*

This has been changed.

> **Reviewer 2**: *P.6: The abbreviations "TCRU" and "TCUU" are not spelled out (although they are spelled out later in the paper, on p. 8*

We have spelled out TCUU early on. We have removed mention of TCRU as now we will only show TCUU trials.

> **Reviewer 2**: *P. 7 (line 23): there is a typo: "auditory" is spelled out as "audiotry"*

This has been fixed.

> **Reviewer 2**: *P. 7 (line 42): the word "trial" is missing between "each…" and "Following…"*

This has been fixed.

> **Reviewer 2**: *If possible, a figure showing an example trial/display in the description of the experimental materials would be helpful for people less familiar with VWP studies.*

Thank you for this suggestion. We have now added a figure highlighting the task.